

unit IP-based Web GUI. It has advanced features such as map display interface for marking locations of all lights in a network over digital maps.”

Technologies to enhance lighting performance

As LEDs become more efficient, performance of a luminaire is judged not just based on light output in lumens, but also the quality of light it emits. Innovations such as tunable liquid lenses have helped create luminaires where beam pattern, focus and colour temperature are adjustable.

GaN substrate LEDs now offer full spectrum output so that the colours are not just mixes of monochromatic RGB LEDs. This allows even more human-centric natural lighting for better productivity during the day and improved sleep quality at night.

Control electronics is improving, too. Low power control gear is now being fitted with power factor correction, higher full-load efficiency and reduced standby consumption. These advances are needed to reduce the lighting load in large buildings by mixing natural and artificial lights for the highest efficiency without harming the environment.

The smart streetlighting system by Maksat Technologies provides streetlight monitoring and management, image monitoring and sensing, and data transmission. ZigBee technology is used to collect data to achieve multi-node low data transmission. It also has environmental sensors that can detect temperature, humidity, carbon dioxide, carbon monoxide, air quality, air pressure and so on. Each streetlight contains a single light control unit and data collect unit attached to the lamp pole. The remote unit gateway is responsible for managing the light and data controls units. One remote unit gateway can manage up to 200 light control and 50 data control units.

Every day new technologies are coming up to enhance the performance of lighting. Organic LEDs are being developed continuously as an alternative to existing LEDs. These are as efficient as high intensity discharge lightings.

Technologies like task optimum/ambient lighting and distribution system lighting, motion sensor/passive infrared motion sensing-based lighting



Fig. 3: Features of smart streetlights
(Credit: Maksat Technologies Pvt Ltd)

are revolutionising the world of lighting. Avnet India has developed a cost-optimum ceiling and wall-mounted PIR motion sensor that can detect motion accurately and automatically switch the lights on/off as per motion detection and visibility. Its supporting features are true motion detection, automatic load on/off, on/off time setting, ambient light sensor and indicating load LED.

Role of lighting in creating a controlled environment for farming

LEDs are slowly being used to control optimum lighting conditions for smart farming. There are technologies like violet LEDs and smart sensors that are helping achieve this. Avnet supports solutions for innovative lighting technologies within the plant-lighting spectrum. In advanced farming technology, plants only use red and blue wavelengths for photosynthesis. Tuned red/blue LED lighting can accelerate crop growth by up to 50 per cent. However, for harvesting, white light is also needed.

Smart agricultural lights with red, deep-blue and white LEDs are becoming standard in many industrial greenhouses. Light spectrum can be adjusted based on the type of crop (flowers, vegetables, etc). Farming is a special case where AI can be used for tuning lighting and watering systems for optimised harvest using data from

humidity and ambient light sensors combined with weather reports.

Prashant Deshmukh, business head lighting - Millennium Semiconductors, says, “Innovations in lighting for indoors and outdoors have different technological requirements. Every designer follows different protocols based on the technology used. Sensors play an important role in achieving controlled and enhanced performance of lightings.”

Threats and areas of security

An important aspect of security is how easily a system can be provisioned and managed. It must be easy to deploy a network from start and then to further maintain this network by adding and removing nodes.

System designers need to follow a set of security principles to protect data and ensure secure data transmission. Any exposed interfaces in applications can be used as attack surfaces and cause harm. In firmware attacks, altered code can modify system behaviour or grant access to sensitive information.

Pointers that need to be kept in mind to achieve safe lighting controls are:

Secure firmware updates. Only authenticated and validated updates can be applied.

Secure physical interfaces and APIs. Data is integrity-protected and authenticated into and out from the module.

Each access grant to gain debug access to a device is unique.

Only authorised usage of APIs is possible as development back doors are blocked.

Robustness spoofing/jamming detection and active countermeasures. Detection of potential attacks on air interfaces is essential.

Secure transport layer. Communication with the server is authenticated and encrypted.

Latest protocols. Bluetooth 5 and Bluetooth Mesh are the new standards coming from Bluetooth SIG.

Self-healing and self-forming ZigBee IEEE 802.15.4, IEEE 802.11 AC Wi-Fi mesh are the standards for Wi-Fi.

DALI, although old, is currently being revamped to bring DALI 2 standard into the modern area. **EFY**